

6.1 #5: 3, 5, 6, 8

6.2 #5: 1, 3, 4, 11

Homework 10

Tyler Troller

3

$\sum_{k=0}^{\infty} \frac{2}{3^k} = 2 \sum_{k=0}^{\infty} \left(\frac{2}{3}\right)^k$ which is a constant multiple of a geometric series where $|r| = \frac{2}{3} < 1$, and thus we have $2 \sum_{k=0}^{\infty} \left(\frac{2}{3}\right)^k$ converges to $2 \left(\frac{1}{1-\frac{2}{3}}\right)$ by Theorem 6.1.6, and this is $2/(1/3) = 6$. Thus, the series converges and it converges to 6. ■

5

$\sum_{k=1}^{\infty} \frac{k^2}{4^k}$ rewrite: $\sum_{k=1}^{\infty} \left(\frac{k}{2^k}\right)^2$, by Example 6.1.12 we have that there exists an N such that $\frac{k}{2^k} < 1$ when $k > N$. Now consider the convergent series in Example 6.1.12. For all k , $\sum_{k=1}^{\infty} \left(\frac{k}{2^k}\right)^2 < \sum_{k=1}^{\infty} \left(\frac{k}{2^k}\right)$, where the terms are always positive for both. Since the series $\sum_{k=1}^{\infty} \left(\frac{k}{2^k}\right)$ converges, and $\sum_{k=1}^{\infty} \left(\frac{k}{2^k}\right)^2 < \sum_{k=1}^{\infty} \left(\frac{k}{2^k}\right)$, we have that $\sum_{k=1}^{\infty} \left(\frac{k}{2^k}\right)^2$ also converges by the comparison test. Note, it is only true that the squared terms are less than the non-squared terms because for all k , each $\frac{k}{2^k} < 1$, and by something previously proved, $|x| < 1 \Rightarrow x^2 < x$. ■

6

$\sum_{k=1}^{\infty} \frac{k}{k^2 - k + 2}$, call this S . Then $S \left(\frac{1/k}{1/k}\right) = \sum_{k=1}^{\infty} \frac{1}{k-1+\frac{2}{k}}$

which is $\geq \sum_{k=1}^{\infty} \frac{1}{k+1}$ since $k \geq 1$ (aka, $-1 + \frac{2}{k} \leq 1 \forall k$) now notice this is $\geq \sum_{k=1}^{\infty} \frac{1}{k+k} = \sum_{k=1}^{\infty} \frac{1}{2k} = \sum_{k=1}^{\infty} \frac{1}{2} \cdot \frac{1}{k}$ which is a constant times a divergent series ($1/k$) proven divergent. So we have shown a series of inequalities that lead to $\sum_{k=1}^{\infty} \frac{k}{k^2 - k + 2} \geq \frac{1}{2} \sum_{k=1}^{\infty} \frac{1}{k}$ which diverges.

Thus, $\sum_{k=1}^{\infty} \frac{k}{k^2 - k + 2}$ diverges as a result. ■

8

$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{\sqrt{k}}$; consider $\sum_{k=1}^{\infty} \left| \frac{(-1)^{k+1}}{\sqrt{k}} \right| = \sum_{k=1}^{\infty} \frac{1}{k^{1/2}}$. Since $k^{1/2} \leq k$ for all k , we have $\frac{1}{\sqrt{k}} \geq \frac{1}{k}$. So, $\sum_{k=1}^{\infty} \frac{1}{k^{1/2}} \geq \sum_{k=1}^{\infty} \frac{1}{k}$ which diverges. So, $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{\sqrt{k}}$ does not converge absolutely because $\sum_{k=1}^{\infty} \left| \frac{(-1)^{k+1}}{\sqrt{k}} \right|$ does not converge. ■

6.2 #5: 1, 3, 4, 11

Homework 10, cont. Tyler Trotter
 #1: $\sum_{k=2}^{\infty} \frac{1}{k \ln(k)}$. Use the Integral Test: A valid test because $\frac{1}{x \ln(x)}$ is a positive, nonincreasing function.

Now consider: $\lim_{b \rightarrow \infty} \int_2^b \frac{1}{x \ln(x)} dx$; let $u = \ln(x)$
 $du = \frac{1}{x} dx$

$$\lim_{b \rightarrow \infty} \int \frac{1}{u} du = \lim_{b \rightarrow \infty} [\ln(\ln(x))] \Big|_2^b = \lim_{b \rightarrow \infty} [\ln(\ln(b)) - \ln(\ln(2))]$$

which goes to infinity as $b \rightarrow \infty$. So, the integral test tells us the series diverges (since it is an if & only statement and we may consider the contrapositive). ■

3: $\sum_{k=1}^{\infty} \frac{k 2^k}{3^k}$. Use the absolute ratio test:
 $r = \lim_{k \rightarrow \infty} \left| \frac{(k+1) 2 \cdot 2^k}{3 \cdot 3^k} \cdot \frac{3^k}{2^k (k)} \right| = \lim_{k \rightarrow \infty} \frac{2(k+1)}{3k}$

which converges to $\frac{2}{3}$ as $k \rightarrow \infty$. So $r = \frac{2}{3} < 1$ which implies the series converges absolutely. ■

4: $\sum_{k=0}^{\infty} \frac{5^k}{k!}$. Use the absolute ratio test:
 $r = \lim_{k \rightarrow \infty} \left| \frac{5 \cdot 5^k}{(k+1)k!} \cdot \frac{k!}{5^k} \right| = \lim_{k \rightarrow \infty} \frac{5}{k+1} \rightarrow 0$

Since $r = 0 < 1$, the ART tells us the series converges absolutely. ■

11: Since b_k is bounded, say $M \in \mathbb{R}^+$ is an upper bound for b_k . Then for all k , $|b_k| < M$. Then $\sum_{k=1}^{\infty} |a_k b_k| = \sum_{k=1}^{\infty} M |a_k| = M \sum_{k=1}^{\infty} |a_k|$. Since $\sum_{k=1}^{\infty} a_k$ converges absolutely, $\sum_{k=1}^{\infty} |a_k|$ converges and a constant multiple of a convergent series is still convergent. So we have $\sum_{k=1}^{\infty} |a_k b_k| \leq M \sum_{k=1}^{\infty} |a_k|$ which implies $\sum_{k=1}^{\infty} |a_k b_k|$ converges and thus $\sum_{k=1}^{\infty} a_k b_k$ converges absolutely. ■